

Introduction to Computer and Computer Science

Homework Assignment #2

Due: 9/30 0:00 am

In this homework assignment, you will practice how to solve some simple numerical problems by writing some simple programs. As you have learned all the related knowledge during the lectures (if, while, for, exhaustive search, and so on...).

Problem #1: Summation of a “string”

Let s be a string that contains a sequence of decimal numbers separated by commas, e.g., $s = "4.99,3.2,5.487,0.18"$. Please write a program that prints the sum of numbers in string s .

Problem #2: Bisection search and Newton-Raphson method

We have already illustrated how to find a cube root via exhaustive enumeration during the class, yet we never demonstrate how to find a cube root via bisection search or Newton-Raphson method. It is time for you to complete our unaccomplished duty. Please write a program that asks users to type in one arbitrary number and print out the corresponding cube root based on the concepts of bisection search and Newton-Raphson method. You should also print out how much iterations (guesses) your program takes for searching the solution based on these two methods as well.

Problem #3: Solving an equation (bonus)

Solving the roots of an equation had been bothering us during our high-school time. Most of the high-school students in Taiwan has dedicated their passion for learning so many techniques in order to find those lovely roots. Fortunately, you had already started to learn some powerful tools for solving the roots of equations since you graduated from high school. This is a great chance for you to forget those dispensable techniques and start to utilize all these efficient tools around you.

Please write a program that can solve the equation as following:

$$\cos x - x = 0$$

You may wonder how to calculate the “cosine” function in Python. Well, try to figure it out by yourself. That’s why this problem is a bonus.

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```
6
7 # =====
8 #
9 # Homework assignment #0
10 # Student: Your name, 你的名字
11 # Student ID: Your student ID, 你的學號
12 # Collaborators: John Doe, Jane Doe, ...
13 #
14 # =====
15 # Your code goes here...
16
17 |
```

Please save your code as a “.py” file, where the file name should follow this format:

108A_hw#?_ID.py

For example,

108A_hw#1_0180506.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “New e3” system. There will be a section for uploading your homework.